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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free Response Questions

20 September 2017

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A Long Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section B Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided.

Answer each part (a) and (b) on a fresh piece of writing paper.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

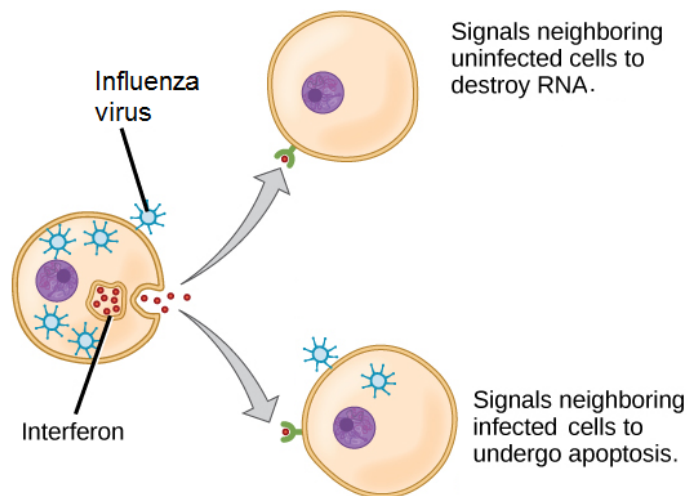
For Examiner's Use	
Section A	
1	/ 25
2	/ 10
3	/ 15
Section B	/25
Paper 3 [75]	
Paper 2 [100]	
Paper 1 [30]	
Paper 4 [55]	
Grand Total [260]	

This document consists of **12** printed pages and **2** blank pages.

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Section A: Long Structured Questions (50 marks)Answer **all** questions in this section.*For
Examiner's
use***Question 1****(a)** Describe how influenza virus infect cells. [4]

Influenza virus infected cells secrete interferon. **Fig. 1.1** shows the functions of interferon.

**Fig. 1.1**

- (b) Using your knowledge, explain why the secretion of interferon is **NOT** considered an adaptive immune response. [2]

*For
Examiner's
use*

- (c) (i) Suggest the importance of destroying RNA in virus infected cells. [2]

- (ii) Interferon induces the expression of RNase L, the enzyme that degrades RNA. This enzyme also degrades mRNA and reduces protein expression. Infected cells are induced to undergo apoptosis due to low protein synthesis and high protein degradation. Name **TWO** molecules required for protein degradation. [2]

Fig. 1.2 shows the binding of interferon to Tyk2 and Jak1 tyrosine kinase receptors which led to the production of RNase L, enzyme which degrades RNA.

For
Examiner's
use

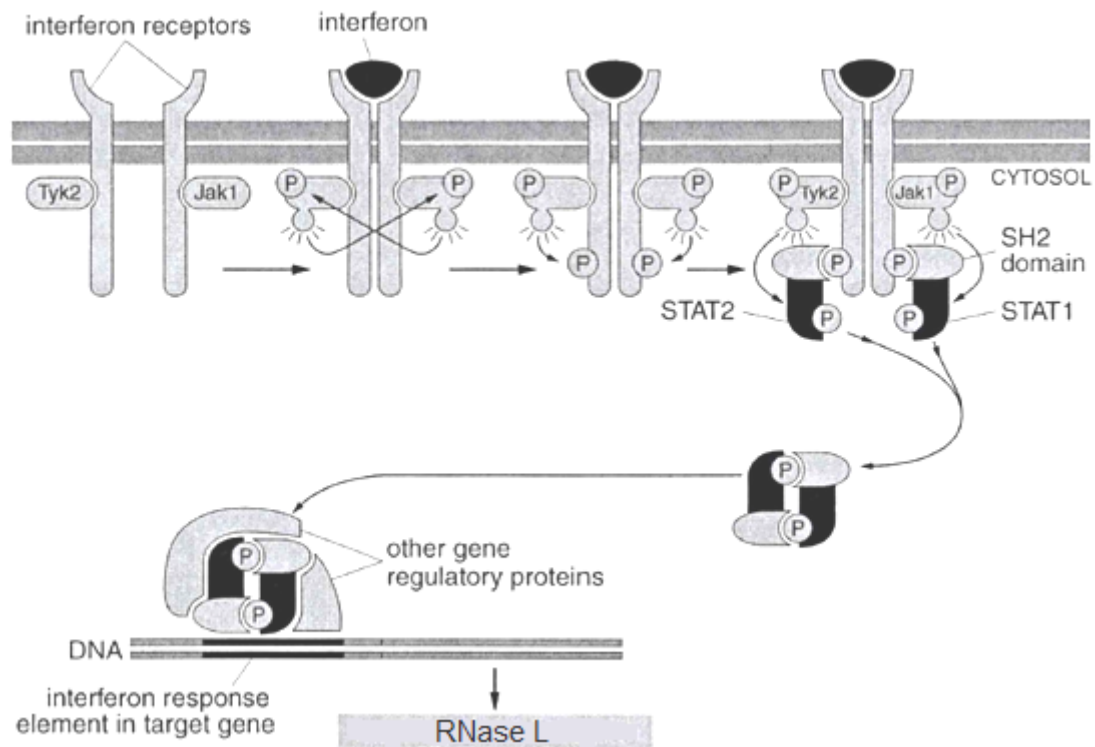


Fig. 1.2

- (d) With reference to **Fig.1.2**, explain how extracellular interferon binding led to the intracellular expression of RNase L. [6]

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(e) Outline how cells develop into cancer cells which divide uncontrollably. [4]

*For
Examiner's
use*

(f) Interferon also has anti-tumour activity. It can increase expression of MHC class I molecules in tumour cells.

(i) Suggest one tumour-specific antigen presented on these MHC class I molecules. [1]

(ii) Explain how interferon lead to cancer cell being destroyed. [4]

Total: [25]

Question 2

Cholera is a disease caused by the bacterium *Vibrio cholera*. The disease symptoms are caused by a toxin, produced by the bacterium.

The cholera toxin is a large globular protein with a mass of 84 kilodalton (kDa) and is composed of two domains, **A** and **B**. Enzymatic domain **A** is made from one polypeptide chain and receptor binding domain **B** is made up of five identical polypeptides.

Fig. 2.1 shows the structure of the cholera toxin.

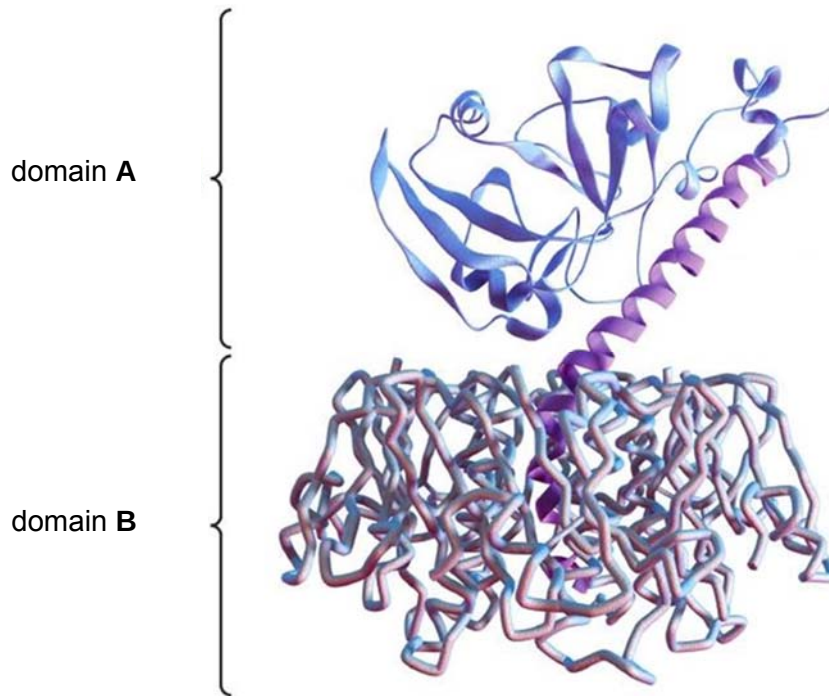


Fig. 2.1

- (a) (i) Describe how the structure of domain **A** of the cholera toxin is maintained. [2]

- (ii) Explain how a globular protein like cholera toxin differs from a fibrous protein such as collagen. [2]

*For
Examiner's
use*

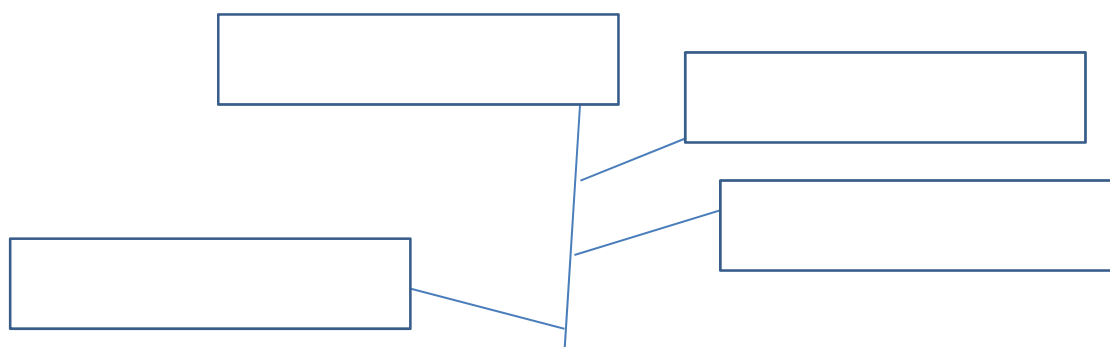
- (b) During an infection, the cholera toxin enters epithelial cells in the human intestine by interacting with receptors on the cell surface membranes. Suggest how the cholera toxin enters epithelial cells. [2]

- (c) There are over 26 strains of *Vibrio cholerae*, which are pathogenic and the cause for cholera disease.

Among *Vibrio* spp., *Vibrio mimicus* seemed to be genetically related to *V. cholerae*, since it was originally reported as an atypical biochemical group of *V. cholerae* strain. A clearer appreciation of the relationship came from the genome-based phylogenetic analysis, where both *V. cholerae* and *V. mimicus* were found to represent two genomically different groups of bacteria that correspond to separate species.

Two *Vibrio* strains, *V. rc341* and *V. rc586*, originally isolated from water samples in Chesapeake Bay, USA represent two novel phyletic lineages within the same clade as *V. mimicus*. These isolates are genetically different from *V. mimicus*, so they have been suggested of being two novel species. However *V. rc341* shows more genomic homology to *V. mimicus* than *V. rc586*.

- (i) Using the information above, complete the phylogenetic tree diagram. [2]



- (ii) Suggest how it is possible for there to be “over 26 strains of *Vibrio cholerae*”. [2]

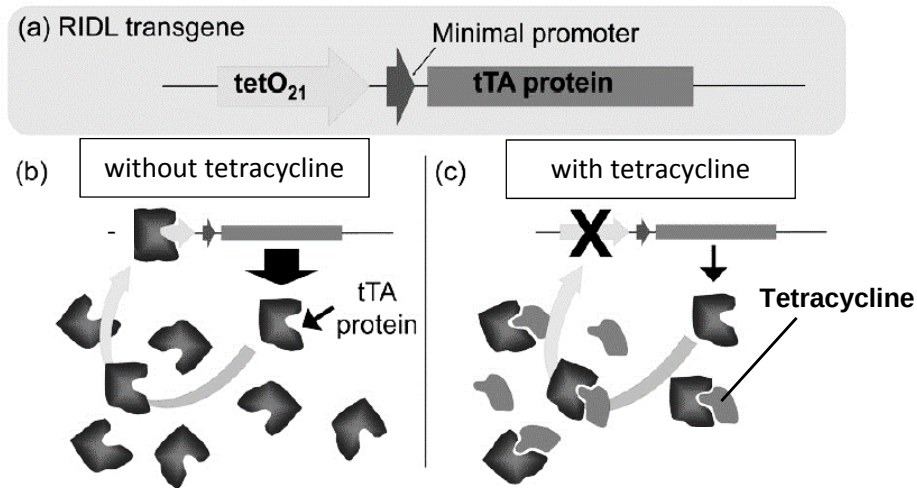
Total: [10]

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Question 3

A new mosquito vector control method involves the dominant lethal genetic system known as (Release of insects carrying a dominant lethal) RIDL. A genetically engineered RIDL system was first demonstrated in the mosquito *Aedes aegypti* which is responsible for the transmission of dengue and malaria.

In RIDL, the mosquito produces activator protein tTA which is toxic to the cell when accumulated in high levels. The regulation of tTA gene is shown in **Fig. 3.1**.

**Fig 3.1**

(a) (i) Define the term *control elements*. [2]

(ii) Contrast the regulation of RIDL and tryptophan operon. [3]

Fig 3.2 shows the numbers of mosquito larvae in traps set up in a control area and in an area where males of the transgenic OX513A strain containing the RIDL system of *Aedes aegypti* were released in Piracicaba county, Brazil in 2015.

Tetracycline is an antibiotic which is generally used in the treatment of infections of the urinary tract, respiratory tract, and the intestines. Tetracycline is not found naturally in the environment.

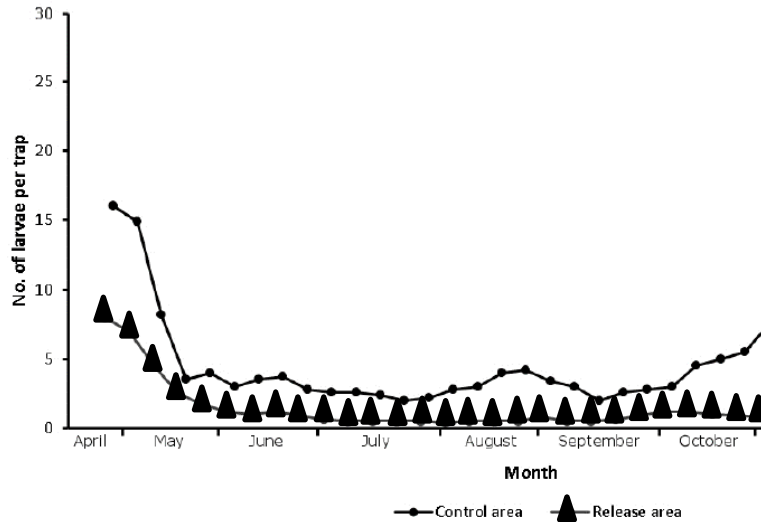


Fig 3.2

- (b) (i) With reference to **Fig 3.1** and **3.2**, describe and explain the effect of releasing transgenic *Aedes aegypti* on the number of mosquito larvae in April. [2]

- (ii) The RIDL mosquitoes are bred to be homozygous dominant for the tTA gene. These mosquitoes have similar mating competitiveness as compared to wild type mosquitoes.

Suggest the benefits of releasing only male homozygous dominant RIDL mosquitoes. [3]

(c) A field study was done to determine the distribution of RIDL mosquitoes after it was released into the wild. To identify RIDL mosquitoes from wild type mosquitoes, samples of mosquito DNA was extracted and tested for the presence of tTA gene.

*For
Examiner's
use*

- (i) Describe the process of polymerase chain reaction which specifically amplifies the tTA gene from the DNA sample. [3]

- (ii) Describe two limitations of polymerase chain reaction in amplifying the tTA gene. [2]

Total: [15]

Section B: Free-Response Question (25 marks)

Answer only one question.

Write your answers on the writing paper provided.

Answer each part (a) and (b) on a fresh piece of writing paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 1

- (a) Describe the various roles of RNA in eukaryotes. [12]
- (b) Describe ATP synthesis in respiration. [13]

Total: [25]

OR

Question 2

- (a) Describe the various bonds and their importance in carbohydrates. [13]
- (b) Describe the differences between Calvin and Krebs Cycles. [12]

Total: [25]

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**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2017
YEAR SIX
H2 BIOLOGY (9744)**

Paper 3

Suggested Answers

Long Structured Questions

1a

- Hemagglutinin bind to sialic acids on host cells.
- The virus enter the host cell by endocytosis forming an endosome.
- Acidic pH in the endosome induces the virus membrane to fuse with the endosome membrane, releasing the nucleocapsid into the cytoplasm.
- Capsid proteins are digested releasing the viral RNA genome into the host cell.

1b

- Secretion of interferon is not a specific response against influenza virus.
- There is no memory formed against influenza virus to give a faster response that is of higher affinity against the virus.
- There are no B and T lymphocytes involved in the removal of influenza virus.

Any 2

1c (i)

- Virus has RNA genome.
- Destroying RNA will prevent virus from replicating / synthesizing viral proteins

1c (ii)

- Proteasome
- Ubiquitin

1d

- binding of interferon to the extracellular binding sites of Tyk2 and Jak1 receptors, causing the receptors to dimerize.
- Tyk2 and Jak1 on the cytoplasmic/intracellular side receptor tails cross-phosphorylate
- Tyk2 and Jak1 adds phosphate group to tyrosine residues on their own tail
- receptor tails are recognised and bind by SH2 domain of STAT1 and STAT2 proteins.
- Tyk2 and Jak1 phosphorylated STAT2 and STAT1
- Phosphorylated STAT1 and STAT2 proteins are activated upon dimerization.
- STAT1 and STAT2 dimer move into the nucleus and bind to interferon response element of RNase L.
- RNase L expression is increased/turned on.

Any 6

1e

- Gain of function mutation in at least one allele
- resulting in overexpression of proteins that stimulates cell division.
- Loss of function mutation in both alleles/copies of tumour suppressor genes
- resulting in no inhibition of cell cycle.

1f (i)

- Telomerase / mutated oncogene

1f (ii)

- Interferon increases expression of MHC class I molecules on the surface of tumour cells, Cytotoxic T lymphocytes with T cell receptors bind complementary to the antigen.
- CD8 glycoproteins bind complementary to MHC class I molecules.
- Any 1 of Cytotoxic T lymphocytes action;
 - o Secretes perforin which forms pores in the cell membrane.
 - o Secretes granzymes which hydrolyses proteins.
 - o Secretes granulysin which induce apoptosis.

2a (i)

- primary structure folds into α helix and β pleated, maintained by hydrogen bonds formed between C=O and NH group
- further folding into a tertiary structure, maintained by hydrogen bonds, ionic bonds, disulfide bonds and hydrophobic interactions between R-groups of amino acid residues

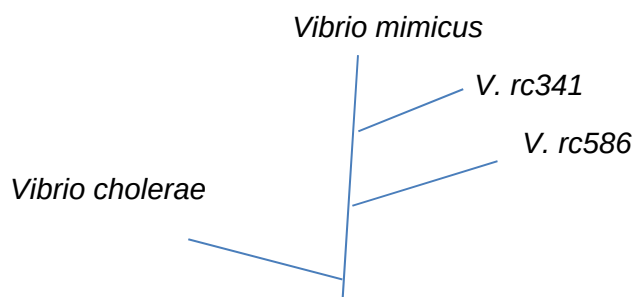
2a (ii)

1. cholera toxin is soluble in water while collagen is insoluble
2. cholera toxin is made up of non-repetitive specific sequence of amino acids, while collagen is made up of repetitive sequence of amino acids
3. cholera toxin is compact in shape while collagen is elongated in shape/forms multimolecular parallel filament to strands/ collagen fibrils and collagen fibres

2max

2b

- receptor binding domain B of cholera toxin binds to receptors on cell membrane
- resulting in a conformational change in receptor
- cholera toxin enters by endocytosis, formation of a vesicle enclosing cholera toxin

2c (i)**2c (ii)**

- Variation within a population is a result of random gene mutations;
- Result in different e.g. cholera toxin, surface antigens,
- in addition, conjugation / transduction / transformation, it is possible that a strain of *V. cholerae* can acquire other forms of variation

max 2

3a (i)

- Control elements are segments of non-coding DNA that help regulate transcription by binding transcription factors
- Promoter, silencer and enhancer are examples of control elements which are bound by RNA polymerase, repressors and activators

3a (ii)

- Negative feedback in tryptophan operon as compared to positive feedback in RIDL
- Trp operon uses repressor proteins to decrease the expression of the operon while RIDL uses activator protein tTA to increase the expression of tTA gene
- In trp operon, tryptophan acts as the co-repressor to activate the repressor protein while in RIDL tetracycline binds and inactivates the regulator protein tTA
- Trp operon is negatively regulated by repressor protein while RIDL is positively regulated by tTA activator protein.

3b (i)

- number of mosquito larvae per trap has dropped from 16 larvae per trap in April to 8 per trap
- As RIDL mosquitoes do not receive tetracycline in the natural environment, RIFL mosquito larvae die

3b (ii)

- Male mosquitoes do not bite and hence they do not transmit diseases
- Releasing homozygous dominant RIDL mosquitoes ensures that all the offspring will carry the RIDL gene
- Male RIDL mosquitoes and wild type male have equal chance to successfully mate with wild type female mosquitoes

3c (i)

- The temperature is increased to 95°C so that the DNA denatures and the hydrogen bonds are broken, separating the double stranded DNA into single stranded DNA;
- The temperature is cooled to 55°C so that primers can anneal to the sequences flanking the tTA gene and hybridize to the single-stranded DNA template via complementary base pairing;
- The temperature is cooled to 72°C so that Taq polymerase recognises the 3' – OH group on the 3' end of the annealed DNA primer and starts the synthesis of complementary DNA strand;

3c (ii)

- Taq polymerase used in most PCR lacks a proofreading mechanism, which results in a relatively high copy error rate
- PCR can only amplify DNA sequences of lengths 0.1 to 5 kb

Free Response Questions**1a** Describe the various roles of RNA in eukaryotes. [12]**mRNA**

1. role in transferring genetic information from nucleus to cytoplasm
2. DNA triplet codes are carried in the form of codons in mRNA
3. Each codon corresponds to one amino acid

tRNA

4. role in carrying the corresponding amino acid to ribosome to match with the codon in translation
5. 3' end binds to corresponding amino acid via covalent bond Attached by to amino acid by aminoacyl tRNA synthetase
6. Contains anti-codon which is complementary to codon on mRNA for translation

rRNA

7. Role in forming ribosome for translation
8. makes up peptidyl transferase which catalysed peptide bond between adjacent amino acid
9. align tRNA and mRNA in ribosome

RNA primer

10. providing 3'OH group for addition of complementary deoxyribonucleotide to growing DNA strand
11. Synthesize by primase

RNA template in telomerase

12. Role in lengthening telomere
13. Expressed in stem cells/gametes

QWC: 2 points from 3 sections

1b Describe ATP synthesis in respiration. [13]

1. ATP is synthesized by substrate level photophosphorylation and oxidative phosphorylation.

Substrate level photophosphorylation

2. ATP is synthesized during glycolysis, in the cytoplasm, and during Krebs cycle in the mitochondrial matrix.
3. 4 ATP / 2 net ATP is synthesized per glucose molecule during glycolysis.
4. In anaerobic respiration, ATP is synthesized only by substrate level phosphorylation in glycolysis.
5. In the Krebs cycle, 2 ATP is synthesized per glucose when succinyl-CoA is converted to succinate.

Oxidative phosphorylation:

6. NAD and FAD are reduced during glycolysis, link reaction and Krebs cycle.
7. Reduced NAD and FAD donates electrons to the electron transport chain on the inner mitochondrial membrane.
8. As electrons are transported along a series of electron carriers of progressively lower energy levels, some energy is used to pump H^+ from the matrix to the intermembrane space, against its concentration gradient
9. This creates a proton gradient across the inner mitochondrial membrane, driving protons to diffuse down its concentration gradient via ATP synthase on the inner mitochondrial membrane.
10. ATP synthase harness the proton motive force for phosphorylation of ADP to ATP, in the mitochondria matrix.
11. O_2 is the final electron carrier of the electron transport chain.
12. 3 ATP is synthesized per reduced NAD and 2 ATP per reduced FAD.

QWC: at least 3 points each from Substrate level photophosphorylation and Oxidative phosphorylation

2a Describe the various bonds and their importance in carbohydrates. [13]

1. Form glycosidic bond by condensation with elimination of one water molecule

Starch/glycogen $\alpha(1\rightarrow4)$ glycosidic bond

2. Form between between anomeric carbon 1 of α glucose and carbon 4 of the other
3. Chain coils helically
4. (10) resulting in a more compact shape for storage

hydrogen bond

5. intra-chain H-bonding between hydroxyl groups helps stabilise helical structure

 $\alpha(1\rightarrow6)$ glycosidic bond

6. Form between between anomeric carbon 1 of α glucose and carbon 6 of the other
7. occurs at branch points
8. (4) resulting in a more compact shape for storage
9. Also, the many branch ends allow a number of amylase to act on starch/glycogen at any one time so it can be easily broken down

Cellulose $\beta(1\rightarrow4)$ glycosidic bond

10. form between β glucose which has 180° rotation of alternating glucose residues
11. forms linear structure of cellulose chain

hydrogen bond

12. Hydroxyl groups project outwards, alternately from both sides of each chain, allowing for the formation of hydrogen bonds between adjacent chains, thus establishing a rigid cross-linking between the chains.
13. Microfibrils can be twisted into a threadlike fibril which further coiled and arranged in larger bundles to form macrofibrils.

QWC: at least 2 points from 3 bonds

2b Describe the differences between Calvin and Krebs cycles. [12]

Marking Point		Krebs cycle	Calvin cycle
1	Location	Mitochondrial matrix	Chloroplast stroma
2	Substrate	Acetyl-CoA and oxaloacetate combines to form citrate	CO ₂ and Ribulose biphosphate (RuBP)
3	Products	Each glucose molecule gives rise to: 6 NADH 2 FADH ₂ 2 ATP 4 CO ₂	For every 3 molecules of CO ₂ that enter the cycle, one triose phosphate / G3P is made
4	Regenerated / Starting material	Oxaloacetate is the starting material that is eventually regenerated	Ribulose biphosphate (RuBP) is the starting material that is eventually regenerated
5, 6	ATP	Produced via substrate level phosphorylation	Used in reduction of glycerate-3-phosphate where energy is required through hydrolysis of ATP
7, 8	Electron carriers / donors	Use NAD ⁺ and FAD for the oxidation of the intermediates of the cycle by serving as electron acceptors	Uses NADPH / reduced NADP ⁺ to reduce glycerate-3-phosphate to triose phosphate by serving as electron donors
9	Overall	Catabolic	Anabolic
10, 11	Role of CO ₂	CO ₂ is released as a result of decarboxylation reactions	Required for carbon fixation. CO ₂ is used to convert Ribulose biphosphate (RuBP) to form an unstable 6C compound that breaks down to form glycerate-3-phosphate
12	Role of O ₂	Occurs only when O ₂ is present	Does not require O ₂

QWC: 5 or more comparisons.